XR Event Planning

Testing Plan for Interactive Prototype 2

1. 3-Sentence Pitch

XR Event Planning is a VR event-planning application that lets users arrange and customize a venue from inside the space. Users can add, move, rotate, and remove furniture, adjust lighting, and change the venue theme through spatial interactions. Following the first evaluation, the prototype now includes clearer furniture feedback, a more visible Remove area, grouped customisation controls, and a Simulator Controls Guide.

2. Testing Objective

IP1 identified difficulties with grabbing furniture, moving unwanted items to the Delete Zone, and remembering simulator controls. Furniture arrangement had the lowest independent completion, at 3/5. Tutor feedback also highlighted unnecessary travel between the customisation areas.

This test examines whether the revised prototype supports independent furniture arrangement and removal, makes the customisation controls easy to find, and provides useful simulator-control support. The focus is on discoverability, control understanding, interaction workflow, and navigation while using the prototype in VR.

3. Assumptions / Unknowns

The test will examine whether:

- The Grab indicator helps users recognise when furniture can be interacted with.
- The redesigned Remove area is easy to identify and use.
- Grouping Remove, Lighting, and Theme makes the customisation workflow easier to follow.
- The Simulator Controls Guide helps users check unfamiliar or forgotten controls without moderator assistance.

4. Testing Methodology

Use task-based usability testing, observation, and think-aloud.

Participants receive three event-planning goals without step-by-step instructions. Observe whether they discover the relevant interactions, complete them independently, use the Guide, or need assistance. Think-aloud comments and post-test questions help explain uncertainty that completion scores alone may miss.

Compare the findings with the issues reported in IP1. Treat this as a comparison between design iterations rather than a controlled test of individual changes.

5. Prototype Description / Requirements

The revised Unity prototype supports:

- Adding a table, chair, and stage.
- Grabbing, moving, and rotating furniture.
- A visible Grab indicator.
- A labelled Remove area with a red spatial effect, positioned near the editing space.
- Brightness adjustment using a virtual lever.
- Theme changes using virtual wands and a pedestal.
- A Simulator Controls Guide accessed through the head-attached interface.
- Remove, Lighting, and Venue Colors/Theme controls grouped along the same general side of the venue.

Review as Guest remains available but is outside this test's focus because it was not a main issue identified in the previous evaluation.

6. Data Collection Method

Record completion separately for furniture arrangement, removal, lighting, and theme, even though lighting and theme belong to the same task.

Outcome	**Score**	**Definition**
Pass	2	Completed independently
Assist	1	Completed after moderator assistance
Fail	0	Unable to complete after assistance

Using the built-in Guide counts as independent use.

Record hesitation, incorrect attempts, difficulty finding controls, Guide use, and any moderator assistance. Note what participants do or say about the Grab indicator rather than assuming that it helped. Keep direct observations separate from later interpretations.

For evaluating the outcomes, use:

- **Validated:** at least 4/5 independent completions.
- **Partially validated:** 3/5 independent completions.
- **Requires further iteration:** 2/5 or fewer independent completions.

Use qualitative evidence to explain these outcomes. Assess Guide usefulness through documented use and feedback, rather than assuming successful completion demonstrates its effectiveness.

7. Participant Data Sheet

Participant ID: _____
Audio-recorded with consent: Yes / No

Interaction	**Outcome: Pass / Assist / Fail**	**Observations and assistance given**
Furniture arrangement		
Furniture removal		
Lighting adjustment		
Theme change		

Guide use: Used / Not used / Not documented
Guide feedback: _____
Response to Grab indicator: _____
Searching or navigation difficulties: _____
Post-test comments: _____

8. Testing Setup

Use a computer running the revised Unity prototype with the VR headset and controllers connected and ready for testing. Prepare the data sheet and an audio recorder where consent is given.

Before each session:

- Reset newly added furniture.
- Restore default lighting.
- Return theme wands to their original positions and clear the pedestal.
- Check the furniture menu, Grab indicator, Remove area, lighting lever, theme interaction, and Guide.

Explain that the prototype is being tested, not the participant. Let participants know the Guide is available, but avoid demonstrating the solutions to the tasks.

Allow participants to attempt the interactions independently. If they become stuck and require help, record the assistance given and score the affected interaction accordingly. Neutral prompts such as “What are you looking for?” should not explain the solution.

9. Testing Process

The timings below are planning estimates, not measured results. Allow approximately six to seven minutes overall, with flexibility for participant feedback.

Introduction: approximately 45 seconds

“Thanks for testing my XR Event Planning prototype. I’ll give you three tasks to try. Please tell me what you’re looking for, what you expect to happen, and anything that feels confusing.

This is a test of the prototype, not of you. You can use the Guide button whenever you need to check the simulator controls. I’ll let you try the tasks yourself before offering help.

Is it okay if I take notes?”

If recording, also ask:

“Is it okay if I audio-record this session for the evaluation?”

Task 1: Arrange the Venue
Approximately 2 minutes

“Imagine you’re preparing this venue for a small event. Add a table, a chair, and a stage and arrange them into a simple layout. Reposition at least one item and change the orientation of at least one object.”

Completion requires all three items to be added, at least one item to be repositioned, and at least one object to be rotated.

Observe furniture-menu discovery, grabbing, moving, rotating, responses to the Grab indicator, Guide use, hesitation, and assistance.

Task 2: Remove Furniture
Approximately 1 minute

“One of the furniture items is no longer needed. Remove it from the venue.”

Completion requires one furniture item to be removed using the Remove area.

Observe whether participants find the area, understand its purpose, and successfully move furniture into it. Record searching, hesitation, and assistance.

Task 3: Customise the Venue
Approximately 1–2 minutes

“Now make the venue noticeably brighter or darker and change it to a different theme.”

Completion requires a brightness change using the lever and a theme change using a wand and pedestal. Score these interactions separately.

Observe discovery of the controls, understanding of the interactions, movement between them, Guide use, and assistance.

10. Post-Test Questions

Approximately 1–2 minutes

1. Was there any point where you were unsure what to do next?
2. Did anything help you understand how to grab or move the furniture?
3. How clear was the Remove area and its location?
4. Were the Remove, Lighting, and Theme controls easy to find and move between?
5. If you used the Guide, was it helpful?
6. What is one thing you would improve?